SRM UNIVERSITY-AP

Department of Computer Science and Engineering

CSE 101 – Fundamentals of Computing and C Programming

CODE2THINK | 30 DAYS CHALLENGE

CHALLENGE 03: SMARTCART

"Calculate. Check. Decide."

Student Name	Registration No.	Challenge No.	Date
J.Jaswanth	AP26110090182	03	11-10-2026

1. Problem Statement

A student is purchasing items from an online store. Read the purchase amount, customer type (1 = Student, 2 = Regular Customer), and delivery distance in kilometres. Determine the discount and delivery charge using conditional statements. Display the purchase amount, discount amount, amount after discount, delivery charge, and final payable amount.

Discount Rules

Condition	Discount
Student + Purchase ≥ ₹2000	15%
Student + Purchase < ₹2000	10%
Regular + Purchase ≥ ₹3000	8%
Regular + Purchase < ₹3000	0%

Delivery Rules

Distance	Delivery Charge
≤ 5 km	₹40
> 5 km and ≤ 15 km	₹70
> 15 km	₹100

2. Objective

Apply if, else if, else, relational/logical operators, and arithmetic expressions to solve a multi-condition decision-making problem.

3. Algorithm – Student Response

Write your algorithm below.

1. Start.
2. Read purchase amount, customer type (1 = Student, 2 = Regular) and distance in km.
3. If type = 1 and purchase >= 2000 set discount 15%; if purchase < 2000 set 10%.
4. If type = 2 and purchase >= 3000 set discount 8%; if purchase < 3000 set 0%.
5. If type is not 1 or 2 print Invalid customer and set discount 0%.
6. discount = (percent / 100) * purchase; amount after = purchase - discount.
7. If distance <= 5 delivery = 40; else if <= 15 delivery = 70; else delivery = 100.
8. Final = after + delivery. Print all values with 2 decimals. Stop.

4. C Program – Student Response

Write your C program below. Use the space provided for your solution.

5. Test Cases & Results

Complete at least 6 test cases. Include the boundary values ₹2000, ₹3000, 5 km and 15 km.

TC	Purchase	Customer	Distance	Discount	Delivery	Final Amount	Result
TC01	2500	Student	20 km	15% 375.00	100.00	2225.00	Pass
TC02	3500	Regular	10 km	8% 280.00	70.00	3290.00	Pass
TC03	3000	Student	25 km	15% 450.00	100.00	2650.00	Pass
TC04	2000	Student	4 km	15% 300.00	40.00	1740.00	Pass
TC05	6000	Regular	16 km	8% 480.00	100.00	5620.00	Pass
TC06	1500	Student	2 km	10% 150.00	40.00	1390.00	Pass

CODE2THINK | CHALLENGE 03
SMARTCART – SUBMISSION FORM

A. Student Details

Student Name	Registration No.	Challenge No.	Date
J.Jaswanth	AP26110090182	03	11-10-2026
Customer Type Used	☐ Student ☐ Regular	**Total / Final Amount**	₹_____

B. Submission Information

Submission	Student Entry
GitHub Repository / Challenge Folder Link	github.com/BOLDGLOBE/novacart-promise-engine
LinkedIn Post Link	linkedin.com/in/jaswanth-joga-2baa74423
60-Second Explanation Video Link	_____

C. What I Learned

Write 4–5 sentences in your own words about what you learned from this challenge and how conditional statements helped you make decisions.

This challenge taught me how if, else if and else let a program choose the right path using real conditions and logical operators. I learned to test boundary values like Rs.2000, Rs.3000, 5 km and 15 km, and my outputs matched every expected result.

D. Student Declaration

I confirm that this submission is my own work and that the program, algorithm, test cases, and explanation are prepared by me.

Student Signature	Date	Faculty Signature	Marks
J.Jaswanth	11-10-2026	_____	_____

E. Evaluation – 10 Marks

Component	Max. Marks	Marks Obtained
Algorithm & Logic	2	_____
C Program & Execution	3	_____
Conditional & Logical Operators	2	_____
Test Cases & Results	1	_____
Submission & Explanation	2	_____

Faculty Remarks

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Think First. Code Better.

Faculty Coordinator: Dr. Yasir Afaq | SRM University-AP